MUHAMMAD ARIEFSYAH

AI Engineer Intern | Machine Learning | NLP | RAG

Jakarta | muhammadariefsyah00@gmail.com | 081317692622
github.com/ArleneNero | linkedin.com/in/muhammad-ariefsyah-524baa152

PROFESSIONAL SUMMARY

Informatics Engineering undergraduate at Universitas Budi Luhur, semester 5 (GPA 3.66/4.00), specializing in AI for Data Text. Experience developing machine-learning and NLP pipelines, evaluating models, and building Retrieval-Augmented Generation (RAG) applications. Team member of the 1st Best Innovation Award at PalComTech Student Conference 2026 and Top 15 Finalist at Smart Move Challenge MRT Jakarta 2026.

EDUCATION

Universitas Budi Luhur | B.Sc. Informatics Engineering | 2024 - Present

Specialization: AI for Data Text. GPA: 3.66/4.00. Relevant courses: Machine Learning, Natural Language Processing, Data Mining.

PROJECTS

Xiaomimi Matcha - In-Memory Hybrid RAG Chatbot (2026)

- Built a React and Express/TypeScript customer assistant grounded in a cafe-specific PDF knowledge base.
- Parsed the document into 22 chunks with 768-dimensional embeddings in an in-memory index.
- Implemented cosine-similarity retrieval with keyword boosting; selects five relevant chunks for Gemini.
- Server-side API credentials, out-of-scope handling, and lexical fallback. Stack: React, TypeScript, Node.js, Express, Google GenAI SDK, Gemini, pdf-parse.

ForeSight - Predictive MRT Crowd Management (Smart Move Challenge 2026)

- Top 15 finalist. Built XGBoost demand forecasts across 13 stations for 15-60 minute horizons.
- Combined transactions, train schedules, weather and event features using BPS-calibrated simulated data.
- Out-of-time evaluation: sMAPE 14.16%, MAE 8.15 passengers per interval; not operational MRT data.
- Developed station/gate crowd-risk modeling, mitigation simulation and historical replay dashboard.

Coordinated Buzzer Detection on X - IGRS Case Study (2026)

- Analyzed 1,264 posts from approximately 932 accounts for possible coordinated behaviors.
- Engineered eight semantic, lexical, temporal and network features; XGBoost achieved approximately 0.90 F1 and 0.98 AUC on hold-out evaluation.
- Used SHAP, DBSCAN, LDA and network analysis to interpret and map potential coordination.

RESEARCH & PUBLICATIONS

Hybrid Evidence Engine and Cascade XGBoost for Early Mental Health Risk Signals

Team research, 2nd PalComTech Student Conference 2026. 1st Best Innovation Award (Computer Science and Informatics).

- Processed 1,899 public X posts with multilingual Sentence-BERT, sentiment and five weighted evidence categories.
- Two-stage Cascade XGBoost: 85.0% accuracy and 0.852 weighted F1 on 100 hold-out cases; rule baseline 58.0%.

- Applied SHAP, safety-first rules and Streamlit monitoring. Statistical-only critical-class recall: 45.5%. Outputs are decision-support signals, not diagnoses.

TECHNICAL SKILLS

Machine Learning & NLP: Python, scikit-learn, XGBoost, Sentence-BERT, TF-IDF, feature engineering, model evaluation, SHAP.

Generative AI & Development: Retrieval-Augmented Generation, LLM API integration, JavaScript, TypeScript, React, Node.js, Express, SQL.

Tools: Git, Jupyter/Colab, Streamlit, Docker, Pandas, Matplotlib, Gephi.

Soft skills and language: Critical analysis, technical documentation, teamwork; Indonesian and English.

AWARDS

- 1st Best Innovation Award, Computer Science and Informatics, 2nd PalComTech Student Conference (30 July 2026).
- Top 15 Finalist, Smart Move Challenge MRT Jakarta (October 2026).